

Course Information

Course Number: *VIST 289*
Course Title: *Special Topics in TouchDesigner and Interactive Multimedia*
Section: *500*
Time: *Tuesday/Thursday 11:10 to 12:50 pm*
Contact Hours: *2 hours lecture and 2 hours lab*
Location: *ARCC 307*
Credit Hours: *3 hours*

Instructor Details

Instructor: *Michael Bruner*
E-Mail: *michaelbruner@tamu.edu*
Office Hours: *By Appointment*

Course Description

Introduction to the fundamentals and practices of real-time interactive multimedia production through the use of the industry leading software, TouchDesigner. Creation of interactive graphics, procedural visuals, data-driven systems and control systems through the use of node based programming. Topics include projection mapping, using sensors, creating user interfaces, interoperability, optimization and rendering. The course will consist of individual assignments to learn fundamentals and techniques with a culminating project at the end of the course.

Course Prerequisites

Prerequisites: Grades of C or better ARTS 104, VIST 115, VIST 210, VIST 275

Course Learning Outcomes

On Completion of the course students will be able to:

- Understand node based programming workflows in TouchDesigner
- Create compelling visuals that incorporate sensor data and interactivity
- Develop organized and optimized node networks
- Display and distribute work across multiple display systems
- Conceptualize how to develop complex modular systems within TouchDesigner

Textbook and/or Resource Materials

No textbook is required for this course.

Students must have a computer that can run TouchDesigner.

These learning modules will be used in class: <https://learn.derivative.ca/>

Grading Policy

Course grades will be calculated based on a combination of weighted scores for projects, homework, and exams. For the semester, the total grade is based on 100 points, which is equal to 100% of your grade. Assessment criteria for group projects will include individual technical and artistic contributions, participation, peer assessment, and overall project success. Assessment criteria for projects will be based upon technical and artistic merit relative to the given criteria. The instructor's qualitative judgment of the student's exercises, projects, and research will include such factors as preparation, conceptualization, technical application, documentation, and aesthetic effectiveness.

Breakdown:

- Individual Assignments / Exercises: 50%
- Final Project: 25%
- Quizzes / Tests: 15%
- Participation: 10%
- Total 100%

To aid in understanding the method by which your project assignments are evaluated, read the following carefully. The actual grading criteria will be related to these categories but will be more specific in nature.

F: 59 and below: *The student work is unresolved; the intentions are unclear and major criteria or goals lack resolution. Presentation is incomplete and/or of poor quality. There is a complete lack of problem solving, creative content and/or technical merit.*

D: 60 -69: *The solution has problems in two or more major areas; skill and design development is marginal or incomplete. The project lacks imagination and/or design potential.*

C: 70 -79: *The student has solved the problem, but the solution lacks depth of understanding; Some program goals are not completely satisfied and the solution contains little promise even though most issues have been addressed.*

B: 80 -89: *The student work shows imagination and potential. Presentation and design content is good; program requirements are fulfilled but in need of more refinement or development. There are no major issues that would require a total redesign of the project.*

A: 90- 100: *The student work has imagination and the solutions to the problems show understanding and thought. The problem solution is highly developed and well presented; the entire assignment shows depth and breadth and is well coordinated. The assignment's potential has been achieved.*

Individual Assignments / Exercises

Individual assignments and exercises will make up a large portion of the course. These will consist of working through TouchDesigner challenges weekly that focus on specific techniques and operators. There will also be assigned videos and tutorials to watch to prepare for the coming week.

Final Project

The final project will consist of an Audio Visual Performance where we will build show files that allow for realtime interaction and control. We will build these projects in groups but students will be assessed as individuals and be responsible for creating a significant portion of the project. There is a strong likelihood we will collaborate with another class on the final project.

Graded Attendance and Participation

The university views class attendance and participation as an individual student responsibility. Students are expected to attend class and to complete all assignments. Please refer to [Student Rule 7](#) in its entirety for information about excused absences, including definitions, and related documentation and timelines.

Your grade will be based upon attendance and active participation in a wide range of class activities. You are expected to be in class from start to finish every day that we have class, (There are exceptions such as health, please see the University Attendance policy: <https://student-rules.tamu.edu/rule07/>). If you do not come to class (this includes virtual attendance), if you leave class early, or are late to class, you risk losing points on your participation grade. Your attendance is worth 5% of your total grade. It is the student's responsibility to inform the instructor of any intended absences. **A student's final grade will be dropped a letter grade for each unexcused absence after 3 unexcused absences (from 4 unexcused absences).** A student, who attends all classes, is well prepared, contributes to class discussion, and demonstrates their engagement with the material and their peers will earn an A for participation.

Excused absences: A list of excused absences can be found in Student Rule 7.1 (<http://studentrules.tamu.edu/rule07/>). Except for absences due to religious Holidays, the student must notify his or her instructor in writing (acknowledged e-mail message is acceptable) prior to the date of absence if such notification is feasible. In cases where advance notification is not feasible (e.g., accident, or emergency) the student must provide notification by the end of the second working day after the absence. This notification should include an explanation of why notice could not be sent prior to the class. If the absence is excused, the instructor must either provide the student with an opportunity to make up any quiz, exam or other graded activities or provide a satisfactory alternative to be completed within 30 calendar days from the last day of the absence.

Excused Absences for Religious Holidays: Texas House Bill 256 (effective 9/1/03) states “An institution of higher education shall excuse a student from attending classes or other required activities, including examinations, for the observance of a religious holy day, including travel for that purpose. A student whose absence is excused under this subsection may not be penalized for that absence and shall be allowed to take an examination or complete an assignment from which the student is excused within a reasonable amount of time after the absence.”

Late Work Policy

- **Weekly assignments** will not be accepted late. Failure to show your work in class on the due date will result in a zero. **Other assignments (including projects, documentation, research)** can be submitted late, but with a penalty of 5 points per day, up to 10 days. Later than that will result in a zero.
- Work submitted by a student as makeup work for an excused absence is not considered late work and is exempted from the late work policy. (See [Student Rule 7.](#))
- Students will be excused from attending class on the day of a graded activity or when attendance contributes to a student’s grade, for the reasons stated in Student Rule 7, or other reason deemed appropriate by the instructor.
- Please refer to [Student Rule 7](#) in its entirety for information about makeup work, including definitions, and related documentation and timelines.
- Absences related to Title IX of the Education Amendments of 1972 may necessitate a period of more than 30 days for make-up work, and the timeframe for make-up work should be agreed upon by the student and instructor” ([Student Rule 7, Section 7.4.1](#)).
- “The instructor is under no obligation to provide an opportunity for the student to make up work missed because of an unexcused absence” ([Student Rule 7, Section 7.4.2](#)).
- Students who request an excused absence are expected to uphold the Aggie Honor Code and Student Conduct Code. (See [Student Rule 24.](#))

Course Schedule

Date	Tuesday	Thursday
Week 1 Aug. 25-29	Lecture: Course Introduction Class: Brief Introductions, Interests and goals Lab: Installing TouchDesigner and Interface Overview	Lecture : TouchDesigner in the wild, Applications and Examples Lab: Build our first patch with TOPS!
Week 2 Sep. 1-5	Lecture: The World of TOPs Lab: Working with Textures, Compositing and Transformation. Operator Overview. EVENT: Michael Out - Recorded Lecture	Lecture: TOPs advanced Lab: Routing video, advanced compositing, creating feedback Assignment: TOP Assignment EVENT: Michael Out - Recorded Lecture

Week 3 Sep 8-12	Lecture: The World of CHOPs Lab: Working with Channels and Parameters. Operator Overview.	Lecture: CHOPs advanced Lab: Timing, audio and patterns Assignment: CHOP Assignment QUIZ #1
Week 4 Sep. 15-19	Lecture: The World of SOPs Lab: Working with Geometry and Rendering. Operator Overview.	Lecture: SOPs advanced Lab: Cameras, PBR and instancing Assignment: SOP Assignment
Week 5 Sep. 22-26	Lecture: The World of Pops Lab: Building particle systems	Lecture: Pops Advanced Lab: Controlling LEDs Assignment: POP Assignment
Week 6 Sep. 29- Oct. 3	Lecture: The World of COMPs Lab: Organization and Output	Lecture: COMPs advanced Lab: Building Interfaces and Controls Assignment: COMP Assignment QUIZ #2
Week 7 Oct. 6-10	Lecture: The World of DATs Lab: Executing code in TD Lecture: Introduction to Final Project Lab: Planning your network	Lecture: Lynn's Class visits Assignment: DAT Assignment
Week 8 Oct. 13-17	FALL BREAK NO CLASS On Tuesday October 14th	Lecture: Performance Mode Lab: Projection Mapping Assignment: Instancing Assignment
Week 9 Oct. 20-24	Lecture: Putting together a show file Lab: Planning your network	Lecture: Intermediate Image Manipulation Lab: Working with Noise, Displacement and Feedback Assignment: Noise System Assignment
Week 10 Oct. 27-31	Lecture: Advanced Interaction - MediaPipe Lab: MediaPipe setup and usage	Lecture: Advanced Interaction - custom sensors Lab: Custom sensors Assignment: MediaPipe Assignment QUIZ #3
Week 11 Nov. 3-7	Lecture: Creating Modular Systems Lab: Working with TOXs	Lecture: Reusable Components Lab: Using the replicator COMP

		Assignment: Replicator & TOX Assignment
Week 12 Nov. 10-14	Lecture: Advanced Visual Generation Lab: Dress Rehearsals - Meet with project teams	Lecture: Weekly Review Lab: Workday Assignment Due: Generative System Assignment
Week 13 Nov. 17-21	Performance Day	Lecture: Weekly Review Lab: Workday QUIZ #4
Week 14 Nov. 24-28	Lecture: TBD Lab: Workday	THANKSGIVING NO CLASS THURSDAY
Week 15 Dec. 1-5	Final Project Presentations	Final Project Presentations
Week 16	Dec. 8 - LAST DAY OF CLASS	

Optional Course Information Items

Canvas (LMS): Information on Canvas is for the sole use of Texas A&M University students enrolled in the class. This online repository contains materials (images, articles, etc.) copyrighted by others that have been included for the purpose of instruction only, under the "fair use" provision of the U.S. copyright laws. Copying these materials to any other site or use of these materials for any other purpose is prohibited. Please log in at <https://canvas.tamu.edu/>

Google Drive: Class slides and other reference materials will be shared via a class Google Team Drive. Assignment submissions will be done via Google Drive as well. You will be added to it in the first week of the semester.

University Policies

Attendance Policy

The university views class attendance and participation as an individual student responsibility. Students are expected to attend class and to complete all assignments.

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including definitions, and related documentation and timelines.

Makeup Work Policy

Students will be excused from attending class on the day of a graded activity or when attendance contributes to a student's grade, for the reasons stated in Student Rule 7, or other reason deemed appropriate by the instructor.

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"The instructor is under no obligation to provide an opportunity for the student to make up work missed because of an unexcused absence" ([Student Rule 7, Section 7.4.2](#)).

Students who request an excused absence are expected to uphold the Aggie Honor Code and Student Conduct Code. (See [Student Rule 24](#).)

Academic Integrity Statement and Policy

"An Aggie does not lie, cheat or steal, or tolerate those who do."

"Texas A&M University students are responsible for authenticating all work submitted to an instructor. If asked, students must be able to produce proof that the item submitted is indeed the work of that student. Students must keep appropriate records at all times. The inability to authenticate one's work, should the instructor request it, may be sufficient grounds to initiate an academic misconduct case" ([Section 20.1.2.3, Student Rule 20](#)).

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You can learn more about the Aggie Honor System Office Rules and Procedures, academic integrity, and your rights and responsibilities at aggiehonor.tamu.edu.

Americans with Disabilities Act (ADA) Policy

Texas A&M University is committed to providing equitable access to learning opportunities for all students. If you experience barriers to your education due to a disability or think you may have a disability, please contact the Disability Resources office on your campus (resources listed below)

Disabilities may include, but are not limited to attentional, learning, mental health, sensory, physical, or chronic health conditions. All students are encouraged to discuss their disability related needs with Disability Resources and their instructors as soon as possible.

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Disability Resources is located in the Student Services Building or at (979) 845-1637 or visit disability.tamu.edu.

Title IX and Statement on Limits to Confidentiality

Texas A&M University is committed to fostering a learning environment that is safe and productive for all. University policies and federal and state laws prohibit gender-based discrimination and sexual harassment, including sexual assault, sexual exploitation, domestic violence, dating violence, and stalking.

With the exception of some medical and mental health providers, all university employees (including full and part-time faculty, staff, paid graduate assistants, student workers, etc.) are Mandatory Reporters and must report to the Title IX Office if the employee experiences, observes, or becomes aware of an incident that meets the following conditions (see [University Rule 08.01.01.M1](#)):

- The incident is reasonably believed to be discrimination or harassment.
- The incident is alleged to have been committed by or against a person who, at the time of the incident, was (1) a student enrolled at the University or (2) an employee of the University.

Mandatory Reporters must file a report regardless of how the information comes to their attention – including but not limited to face-to-face conversations, a written class assignment or paper, class discussion, email, text, or social media post. Although Mandatory Reporters must file a report, in most instances, a person who is subjected to the alleged conduct will be able to control how the report is handled, including whether or not to pursue a formal investigation. The University's goal is to make sure you are aware of the range of options available to you and to ensure access to the resources you need.

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Students wishing to discuss concerns in a confidential setting are encouraged to make an appointment with [Counseling and Psychological Services \(CAPS\)](#).

Students can learn more about filing a report, accessing supportive resources, and navigating the Title IX investigation and resolution process on the University's [Title IX webpage](#).

Statement on Mental Health and Wellness

Texas A&M University recognizes that mental health and wellness are critical factors that influence a student's academic success and overall wellbeing. Students are encouraged to engage in healthy self-care by utilizing available resources and services on your campus

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Students who need someone to talk to can contact Counseling & Psychological Services (CAPS) or call

the TAMU Helpline (979-845-2700) from 4:00 p.m. to 8:00 a.m. weekdays and 24 hours on weekends. 24-hour emergency help is also available through the National Suicide Prevention Hotline (800-273-8255) or at suicidepreventionlifeline.org.

Use of AI in the Classroom

AI tools are allowed to generate content (text, video, audio, images) for any work (assignments, activities, responses, etc). Since those will be part of your evaluation in this course, any student work submitted using AI tools should clearly indicate what work is the student's work and what part is generated by the AI. When AI is being used for class projects and/or assignments, students **MUST** cite which AI application they used, and describe how it was used in their assignment. If any part of this is confusing or uncertain, please reach out to your instructor for a conversation before submitting your work.